

Shruti Narmeti - Quant Project

1. Introduction

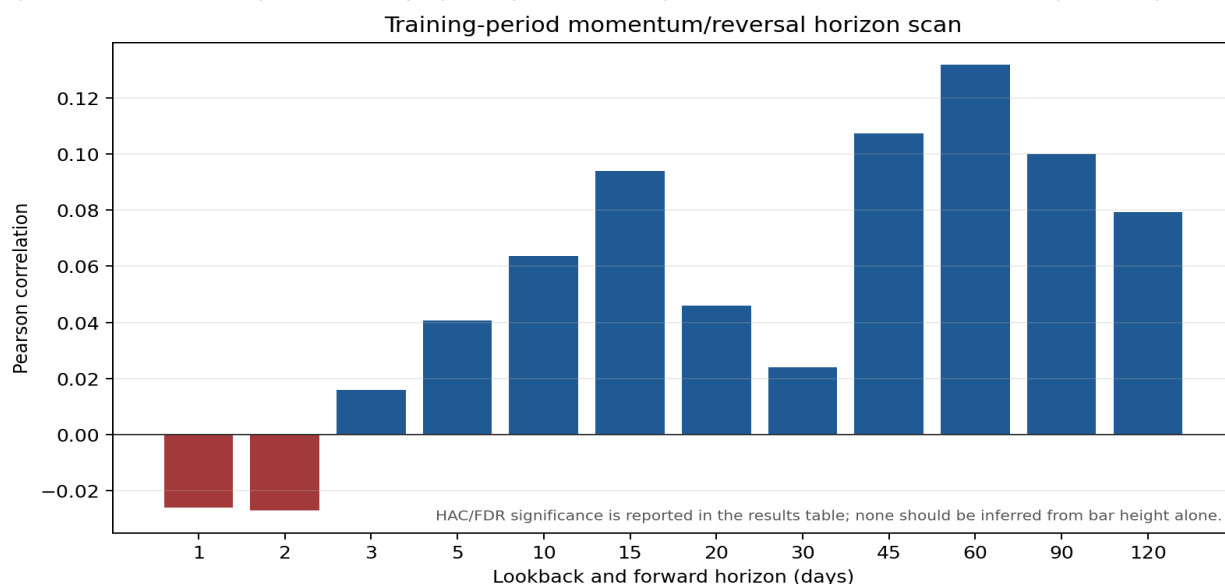
This project studies whether recent BTC/USD price movement contains a useful momentum or reversal signal. The final strategy is a single-asset, long-only Bitcoin momentum rule selected through a train/validation/test research design. The goal is not to prove a permanent trading edge, but to test a disciplined hypothesis with transaction costs and held-out performance.

2. Data Collection and Preprocessing

The data is Bitstamp BTC/USD one-minute OHLCV history from 2012-01-01 through 2025-01-07. The minute records are validated, duplicate timestamps and missing values are rejected, incomplete calendar days are removed, and the clean data is converted into complete daily open, high, low, close, and volume observations. The formal analysis begins on 2015-01-10 to avoid the earliest, least-liquid BTC regime.

3. Momentum and Reversal Research

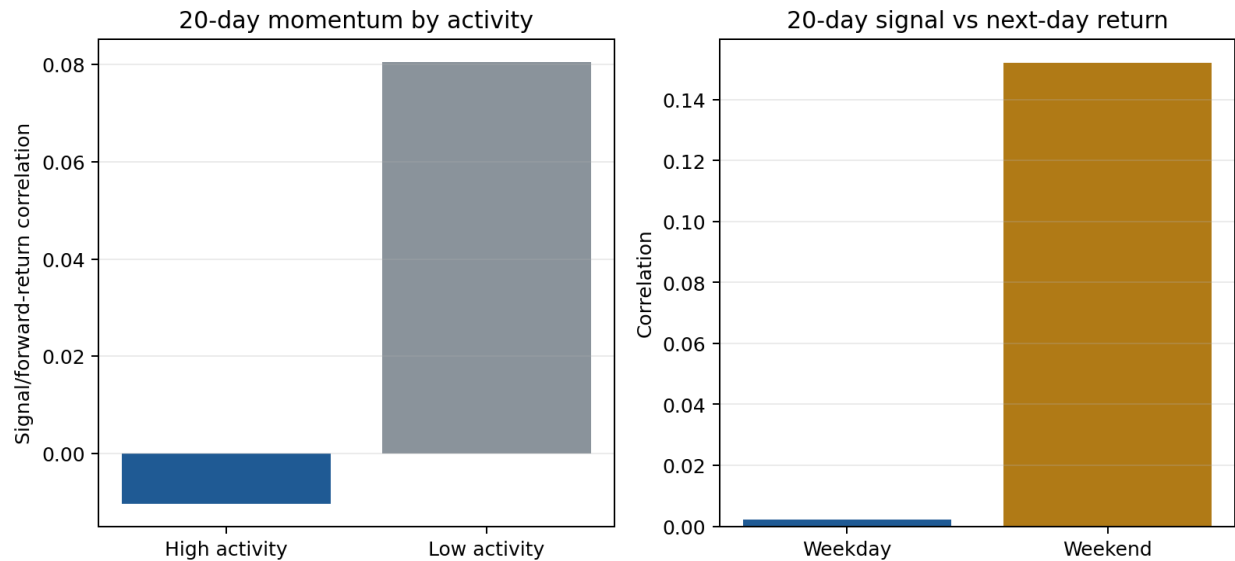
The main research test compares past log returns with future log returns across 1, 2, 3, 5, 10, 15, 20, 30, 45, 60, 90, and 120 day horizons. Positive correlations suggest momentum; negative correlations suggest reversal. Because overlapping multi-day returns are not independent, the project reports HAC-adjusted inference and false-discovery-rate adjustments.



Training-period results were suggestive but not definitive: medium and long horizons were directionally positive, while no horizon remained significant at the 5% level after the final statistical corrections.

4. Activity and Seasonality Tests

The project also asks whether the signal behaves differently during higher-volume periods or when trades are entered on weekends versus weekdays. These tests were included because crypto trades continuously and market participation can vary by time and activity level. The evidence did not strongly support using either feature as the final trading filter.



5. Strategy Selection and Backtest Design

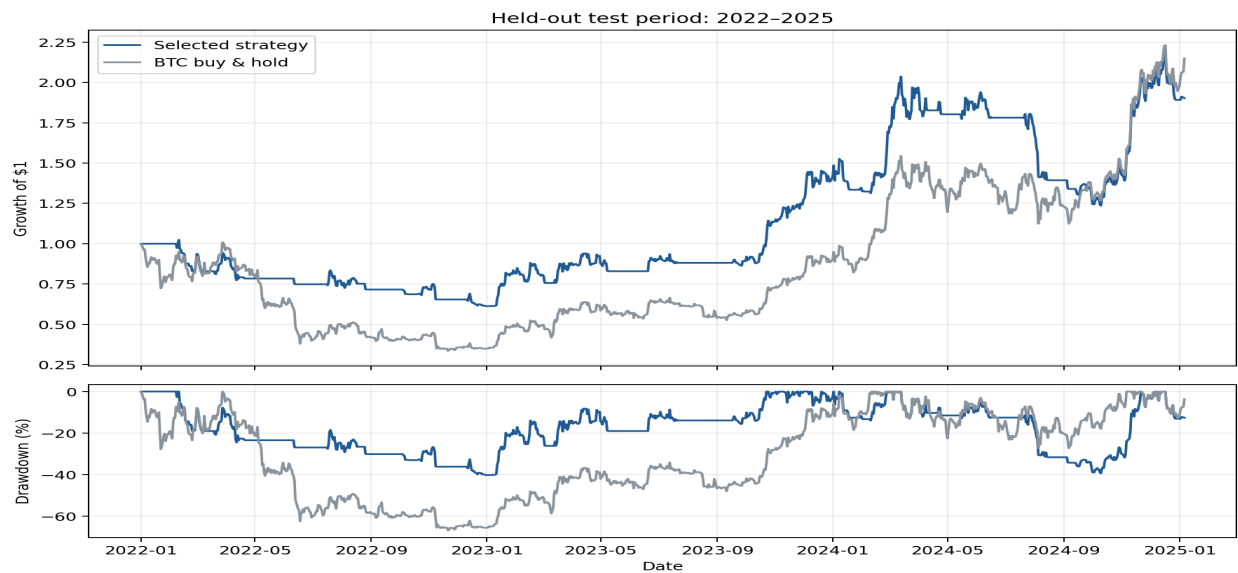
The sample is split into training, validation, and held-out test periods. Candidate rules vary the momentum horizon, long-only versus long/short positioning, and the activity filter. The selected rule is chosen using validation Sharpe ratio only, with lower turnover used as a tie-breaker. The final test period is not used for selection.

Period	Dates	Purpose
Training	2015-01-10 to 2019-12-31	Research signal family
Validation	2020-01-01 to 2021-12-31	Choose final configuration
Test	2022-01-01 to 2025-01-06	Evaluate selected rule

The selected strategy uses a 30-day lookahead, long-only positioning, no activity filter, and a 20 bps market-order cost. If the trailing return is positive, the strategy holds Bitcoin the next day; otherwise it remains flat. The one-day shift prevents look-ahead bias.

6. Performance

Metric	Selected strategy	BTC buy and hold
Total return	90.3%	119.3%
CAGR	23.8%	29.7%
Annualized volatility	36.7%	53.7%
Sharpe ratio	0.76	0.75
Maximum drawdown	-40.2%	-66.8%
Annualized alpha	9.1%	N/A
Beta	0.47	N/A



The selected strategy was profitable during the held-out period and produced lower volatility and a smaller drawdown than holding Bitcoin continuously. It earned less total return than buy-and-hold, but had a slightly higher Sharpe ratio. Its beta of 0.47 shows meaningful BTC exposure, so the strategy is directional rather than market neutral.

7. Trading Costs and Robustness

Scenario	Cost	Total return	CAGR	Sharpe	Max drawdown
zero_cost	0 bps	131.6%	32.1%	0.94	-36.6%
limit_order_7bps	7 bps	116.2%	29.1%	0.88	-37.6%
market_order_20bps	20 bps	90.3%	23.8%	0.76	-40.2%

8. Statistical Significance

The strategy's daily returns are regressed against BTC daily returns using HAC standard errors. The test-period alpha estimate is positive, but it is not statistically significant at the final threshold. This keeps the conclusion appropriately conservative: the backtest is useful evidence for a defensive momentum rule, not proof of a guaranteed edge.

9. Conclusion

The final project demonstrates a reproducible cryptocurrency momentum study. A 30-day long-only Bitcoin rule was selected without using the held-out test period, remained profitable after 20 bps trading costs, and reduced volatility and drawdown relative to buy-and-hold. The main limitation is that it studies only BTC/USD and remains exposed to the Bitcoin market. Future work should expand to multiple assets, cross-sectional momentum, reversal baskets, funding-rate data, and rolling walk-forward validation.